Unit: 6 Unit 8: Quadratics
Topic: Systems of Quadratic Equations

Name: ________________
Date: ____________

Challenge (Habanero)

Q1. Solve the system: $x^2 + y^2 = 25$, $x^2 - y^2 = 9$
(A) $x = 4, y = 3$
(B) $x = -4, y = -3$
(C) $x = 2, y = 1$
(D) $x = -2, y = 1$
(E) $x = 1, y = 2$

Q2. Graph the system: $y = x^2 - 2x + 1$, $y = x^2 + 2x - 1$
(A) Two intersecting parabolas
(B) Two non-intersecting parabolas
(C) One parabola and one line
(D) Two lines
(E) One circle

Q3. In a gaming tournament, a player's score S is given by $S = x^2 + 2y^2$, where x is the number of wins and y is the number of losses. If $S = 50$, find x and y.
(A) $x = 5, y = 2$
(B) $x = 2, y = 5$
(C) $x = 7, y = 1$
(D) $x = 1, y = 7$
(E) $x = 3, y = 4$

Q4. A tech company has two data plans: $y = 2x^2 + 10x + 5$ and $y = x^2 - 5x - 3$. Find the point of intersection.
(A) $x = -2, y = 3$
(B) $x = 2, y = -3$
(C) $x = 1, y = 2$
(D) $x = -1, y = -2$
(E) $x = 3, y = 4$

Q5. Solve the system: $x^2 - 2y^2 = 1$, $x^2 + 2y^2 = 9$
(A) $x = 2, y = 1$
(B) $x = -2, y = -1$
(C) $x = 1, y = 2$
(D) $x = -1, y = -2$
(E) $x = 3, y = 4$

Q6. A gaming console has a rating system given by $R = x^2 + y^2$, where x is the number of hours played and y is the number of achievements. If $R = 100$, find x and y.
(A) $x = 8, y = 6$
(B) $x = 6, y = 8$
(C) $x = 10, y = 0$
(D) $x = 0, y = 10$
(E) $x = 5, y = 5$

Q7. Graph the system: $y = x^2 - 3x - 2$, $y = x^2 + 3x + 2$
(A) Two intersecting parabolas
(B) Two non-intersecting parabolas
(C) One parabola and one line
(D) Two lines
(E) One circle

Q8. A tech startup has two apps with ratings $R_1 = x^2 + y^2$ and $R_2 = x^2 - 2y^2$. Find the point of intersection if $R_1 = R_2$.
(A) $x = 2, y = 1$
(B) $x = -2, y = -1$
(C) $x = 1, y = 2$
(D) $x = -1, y = -2$
(E) $x = 0, y = 0$

Open Ended Questions (FRQ)

Q9. Solve the system: $x^2 + y^2 = 16$, $x^2 - y^2 = 4$. Show all work and write the final answer in the format (x, y).

Q10. A gaming company has a scoreboard system given by $S = x^2 + 2y^2$, where x is the number of wins and y is the number of losses. If $S = 50$, find x and y and write a brief explanation of how you arrived at your answer.

Chef's Tips

- Tip 1: Encourage students to use graphing calculators to visualize the systems of quadratic equations.
- Tip 2: Remind students to check their work by plugging their solutions back into the original equations.
- Tip 3: Emphasize the importance of showing all work and using proper mathematical notation.